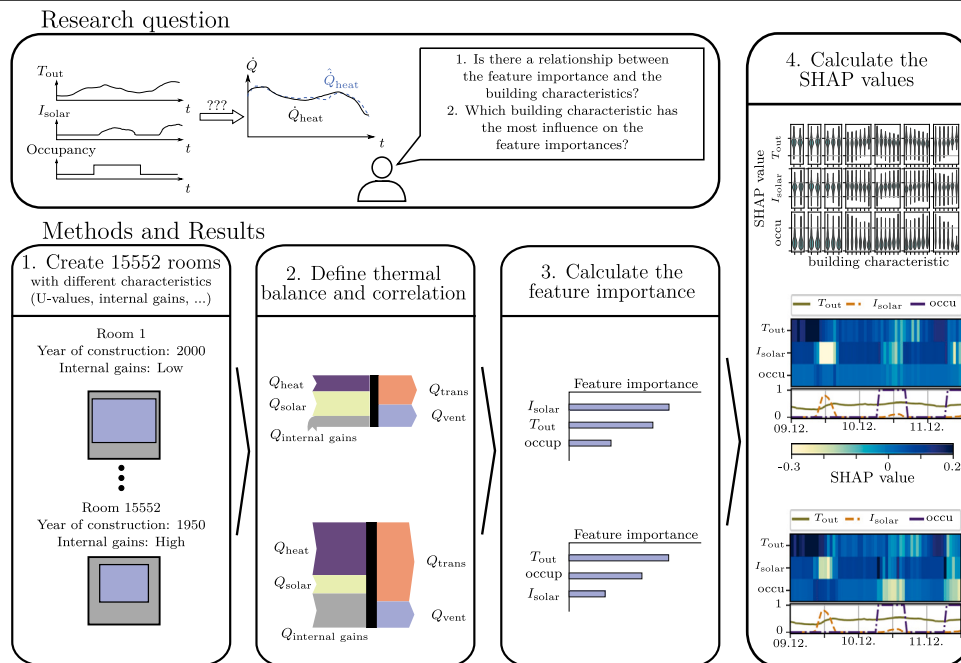


# Relationship between feature importance and building characteristics for heating load predictions

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## GRAPHICAL ABSTRACT



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## ABSTRACT

The use of machine learning in building technology has become increasingly important in recent years. One of the applications is heating load prediction, which enables demand-side flexibility. Most studies consider the heating load prediction without sufficient context with the existing building characteristics. For an accurate load prediction, suitable features have to be selected according to their importance, the feature importance (FI). The scope of this paper is to investigate whether there is a relationship between the building characteristics and the FI and if so, how strong this relationship is. Additionally, an analysis has been conducted to determine which building characteristic have the most significant impact on FI. For this purpose, a full factorial design of a room with six different building characteristics is carried out. In total, the heating load is calculated for 15552 room variants. The thermal balance, correlation, random forest FI, permutation FI and SHapley Additive exPlanations (SHAP) values are calculated for these different rooms. The local SHAP values were

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used to explain the model. These values also provide insight into the interaction of individual features with the heating load. For most variants, the outdoor temperature had the highest FI. It is investigated which building characteristics have the greatest influence on the thermal balance, correlation, FI and SHAP values. A relationship was found between the proportion of thermal balance, the correlation between the features and the label as well as the FI. The greatest association with the thermal balance characteristics was found for the SHAP values. The study shows the systematic relationship between building characteristics and FI. Therefore, FI should always be considered in the context of building characteristics.

## Nomenclature

### Greek Symbols

$\phi$  Shapley value

### Latin Symbols

$\dot{Q}$  Heat flux (W)  
 $\dot{q}$  Heat flux density ( $\text{W m}^{-2}$ )  
 $A$  Area ( $\text{m}^2$ )  
 $D$  data set (–)  
 $e$  Error (–)  
 $f$  Model (–)  
 $g$   $g$ -value (–)  
 $I$  Irradiance ( $\text{W m}^{-2}$ )  
 $i$  Player (–)  
 $K$  Number of repetitions (–)  
 $L$  Error metric (–)  
 $N$  Set of Players (–)  
 $n$  Running number (–)  
 $Q$  Heat (W s)  
 $r$  Pearson Correlation Coefficient (–)  
 $r_s$  Spearman's Rank Correlation Coefficient (–)  
 $S$  Coalition of Players (–)  
 $T$  Temperature (K)  
 $t$  Time (s)  
 $v$  Characteristic function (–)  
 $x$  Feature (–)  
 $y$  Label (–)

### Subscripts

ext exterior  
 IG Internal gains  
 out Outside  
 perm Permutation  
 sol Solar  
 trans Transmission  
 vent Ventilation  
 wall Wall  
 win Window

installed capacity of heat pumps is more than 1000 GW and will increase to 2600 GW by 2030 according to the Announced Pledges Scenario (APS) [1]. The electricity required for the heat pumps is provided by an increasing share of renewable energy in the electricity grid. The global renewable electricity capacity will grow by more than 60% between 2020 and 2026, reaching more than 4800 GW. The increase in renewable electricity capacity is largely due to photovoltaics and wind. These are fluctuating electricity generation sources [2]. Heat pumps offer a good opportunity for demand-side flexibility. The demand-side flexibility capacity of heat pumps in the EU increases by a factor of more than three between 2021 and 2030 [2]. Heat pumps will have a share of about 12% of the total flexibility resources in the EU in 2030 [1].

In order to use demand-side flexibility, heating load predictions with a good prediction accuracy are necessary. Data-driven approaches, which have recently gained a lot of attention, are a useful way to predict the heating load. There are two types of data-driven approaches: statistical methods and machine learning methods [3]. Data-driven approaches are black-box models in which the relationship between the input variables, and the output variables respectively the predicted variables, is learned using training data. In contrast to grey-box models with a physical model, it is possible to dispense with the often complex and time-consuming modelling of the building. For the application of machine learning (ML) models for heating load prediction, it is important to select suitable input variables, the so-called *features* in ML. The suitable features are determined by feature selection. By selecting suitable features, the training time and model complexity can be reduced and in some cases even the prediction accuracy can be increased [4]. The selected features and the accuracy of the prediction should be considered in relation to the characteristics of the building. Since the characteristics of the building are present implicitly in the training data. Another property of black-box approaches is the low interpretability of the results. The interpretability decreases with increasing model accuracy [5]. Model explainability is used to explain the results of the model. It describes how the model gets from the inputs to the results. There are two different types of model explainability model-specific – depending on the used ML model – or model-agnostic, meaning without dependence on a model and only depending on the data set.

Previous studies have almost always considered the feature importance (FI) for heating load predictions without taking the building characteristics into account. There is currently a research gap as no comprehensive and systematic investigation of the influence of the building characteristics on the FI for heating load predictions has been carried out. In the scope of this work, a parameter study with a generic room model is used to simulate 15 552 different variations with various building characteristics and internal loads as well as air change rates in order to determine the hourly heating load for one year. For these 15 552 data sets, the thermal balance of the building over one year is evaluated as well as the FI and model explainability are calculated. FI is determined using random forest and permutation. The model explanation is carried out using SHapley Additive exPlanations (SHAP) values. The aim of the work is to establish a connection between the characteristics of the room, the FI and the model explainability. The question is whether the characteristics of the building are reflected in the results and if this relationship allows a better interpretation of the model. In addition, the importance of the individual features for the prediction quality will be determined depending on the building type.

## 1. Introduction

Heating load predictions are a key technology for the energy transition. Demand-side flexibility can be achieved through the use of heating load predictions. In the coming years, the sector coupling between the electricity sector and the heating sector will increase strongly. Worldwide, 42% of the space heating demand in 2021 was covered by natural gas and 15% by electricity. In the coming years, the share of gas will drop significantly and the share of electricity will rise sharply. This is due to the rising number of heat pumps. In 2021, the global

Three types of sources for building energy prediction uncertainties can be distinguished, the human factors, the building factors and the weather factors. The weather factors have a large influence [6]. When considering the forecast deviation of the weather factors, it can be seen that the solar irradiance prediction has a significantly higher mean absolute percentage error than the ambient temperature prediction [7]. The uncertainty of the input variables leads to uncertainty in the thermal load prediction [8]. Based on the importance of the input variables – the FI – conclusions can be drawn about the prediction accuracy. For example, if the FI of solar irradiance and outdoor temperature is the same, the solar irradiance is probably involved to a greater proportion in the heating load prediction error, because its input data have a greater uncertainty. The main contributions of this work are as follows:

1. Describes the relationship between the building characteristics, the thermal balance, the correlation between the features and the heating load, the FI determined by random forest and permutation, as well as the SHAP values.
2. Define the FI depending on the building characteristics.
3. Determine the influence of the building characteristics on the variation of the thermal balance and the FI.
4. Show the local SHAP values over the time course for different building types depending on the feature values.

The paper is divided into six sections. Section 1 contains the thematic introduction to the topic and explains why heating load predictions are necessary and why they should be considered in the context of the room characteristics. In Section 2 the state of the art and already conducted investigations of feature selection and model explainability in the context of energy and heating load predictions are presented. The description of the methodology and the used data is given in Section 3. The results of the parameter study, the thermal balance, the correlation, the FI and the model explanation are presented in Section 4. Section 5 contains the conclusion with questions for future research.

## 2. Related work

The field of heating load prediction using machine learning has been gaining increasing importance in recent years. The number of papers on Explainable Artificial Intelligence in energy and power system applications [9] has increased significantly in recent years. Sakkas and Abang [10] determined the Spearman's and Kendall's coefficients for real measured data between the hourly heating load of a district heating system and various features. The strongest correlation was between temperature and the heating load. Whereas the wind speed and the rain index have only a low correlation. Chen et al. [11] evaluated 25 papers for building energy prediction with different prediction time horizons. The outdoor dry bulb temperature was the most frequently used feature with 18 uses, followed by outdoor relative humidity (13) and solar radiation (8). In some studies such as Fan et al. [12], a large number of features, among others meteorological features in this case 12, are used for the calculation of energy consumption. It was found that a large number of meteorological features have only a minor importance. In predicting the heating load for a real district heating network [13], the past load, the outdoor temperature and the time index were determined as the most important features.

Kapetanakis et al. [4] selected appropriate features for data-driven prediction models for the heating and cooling load. Therefore they used the Pearson and Spearman correlation. Six different reference buildings each of 16 different locations were considered. The ambient temperature had a strong correlation in almost all cases, while the other input factors like zone air temperature, zone and air humidity and the solar radiation correlates variously with the thermal load. Depending on the building type and the climate zone. The prediction accuracy of an artificial neural network was examined with a reduced feature selected on their correlation with the thermal load. In the majority of

the cases the accuracy had the same level and the execution time was decreased. In the work by Chaganti et al. [14], the heating and cooling load for different ML models was investigated for a data set with 768 buildings with eight different features. In Chaganti et al. the heating load is defined as the amount of energy for an unspecified longer period of time, whereas in this study the heating load is the hourly load. The features used were glazing area, orientation, height, relative compactness, roof area, surface area, and wall area. The relative compactness, surface area and wall area had the greatest influence on the determination of the heating and cooling load. In the work of Ibrahim et al. [15], the annual heating and cooling load was determined for 3840 typical buildings in Saudi Arabia. The features with the highest importance for the multilayer perceptron were the  $U$ -values of the window and the outer wall, the window-to-wall ratio, whereas the floor height and the building size had only a minor influence.

Chung and Liu [16] created a data set of over 2000 buildings in five different climate zones and the constraints of U.S. Department of Energy reference buildings. For these buildings, the annual heating and cooling demand was calculated and the importance of the nine features was determined using the standardised regression coefficient (SRC), local interpretable model-agnostic explanations (LIME) and SHAP values. The aim of the study was to reduce the number of features required while maintaining or only slightly reducing the prediction accuracy. In the investigation of the SRC, the global solar radiation and the dry-bulb temperature are the variables with the highest influence on the heating load. For the LIME values, the dry-bulb temperature also had the greatest influence and the largest variation. For the SHAP values, the dry-bulb temperature had the greatest influence on the heating load and time also had a high influence. The importance of the SRC, LIME and SHAP values were ranked. The SRC has a similar tendency of importance as the physical phenomena of the heat transfer. Whereas the LIME and SHAP values have a tendency to deviate from the physical understanding. The SHAP value was found to be the most credible analysis method for determining the importance of features for heat and cooling load prediction. In the work of Deb et al. [17] the best retrofit strategies for buildings were identified using the permutation FI and SHAP values. It was found that it is promising to choose the retrofit strategy based solely on the FI. Wu et al. [18] shows for the heat and cool load prediction that the local SHAP values for summer and winter and thus the importance of the feature differ strongly. The connection between the features and labels could be well mapped and visualised with the SHAP values. The local SHAP values are consistent with the objective understanding and increase the transparency of the prediction model and the credibility of the results.

The impact of the same features on building energy differs per building. Deciding on the selection of features require a good understanding of the building physics and the specific building itself. This process may require the involvement of expert knowledge and bespoke modelling, which will drive up development costs [19]. For this reason, this paper examines the relationship between FI and building characteristics. It also examines which building characteristics have the strongest influence on the FI.

## 3. Methods and materials

### 3.1. Evaluation metrics

In the scope of this study, the *Pearson correlation coefficient* (PCC)  $r$  (Eq. (1)) and the *Spearman's rank correlation coefficient* (SRCC)  $r_s$  (Eq. (2)) are used to evaluate the correlation between two variables. The PCC describes the measure of linear correlation between two variables  $(x, y)$ . The SRCC assesses the monotonic relationship between two continuous or quantitatively discrete variables independent of linearity. The  $r_s$  is calculated using the pairwise distances (between  $x$  and  $y$ ) of the ranks  $d$  and the number of samples  $n$ . Both evaluation variables can assume values between  $-1$  and  $1$ . A positive value means a rising trend

and a negative value means a falling trend. The closer the correlation is to the value  $\pm 1$ , the stronger the correlation is. The classification of the correlation strength into fixed ranges with cutoff points is problematic because the interpretation of the correlation coefficients depends on the underlying data and the field of application [20].

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}} \quad (1)$$

$$r_s = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)} \quad (2)$$

### 3.2. Multivariate analysis of variance

The *multivariate analysis of variance* (MANOVA) is a statistical analysis method that analyses the influence of several independent variables on more than one dependent variable. The linear combinations of the different dependent variables are examined for significant differences in the mean value. The MANOVA is a further development of the *Analysis of variance* (ANOVA). In the ANOVA, there is only one independent variable, whereas in MANOVA there are several independent variables. To evaluate the relationship between the independent and dependent variables, the Wilks Lambda (WL), among others, can be used. The WL indicates how strongly the independent variable divides the dependent variable into individual groups. A lower value means a greater discrimination ability of the independent variables to the dependent variables.

### 3.3. Feature selection

The objective of feature selection is to determine suitable features or subsets of features for the prediction model. A proper selection of features can, for example, reduce the complexity of the model, shorten the training time and increase the quality of the prediction. Furthermore, the interpretability of the model can be enhanced by an appropriate choice of features. Depending on whether label information is available or not, feature selection techniques can be divided into three families: the *supervised methods*, the *semi-supervised methods* and the *unsupervised methods* [21]. In the scope of this work, the labels are known, so they are *supervised methods*. Three different approaches to feature selection can be distinguished the *filter methods*, the *wrapper methods* and the *embedded methods* [22]. Furthermore, feature selection can be distinguished between model-specific and model-agnostic evaluation methods. With model-agnostic methods, the evaluation results are independent of the selected learning algorithm, whereas the results of model-specific methods are dependent on the selected learning algorithm. Fig. 1 shows an overview of selected methods of model selection and model explanation.

Filter methods determine the importance of features independently of the used learning algorithm (model-agnostic). When using filter methods, single features (univariate feature filters) or feature subsets (multivariate feature filters) can be used. An example of a univariate filter variant is the determination of the correlation between the features and the label [24]. For this purpose, the PCC, SRCC (Eq. (1), (2)) are used in the scope of this work. The correlation for each feature is determined, sometimes ranked, and features with weak correlation can be removed. Filter methods have a low computing time. However, the correlation methods only examine the relationships of individual features to the label. The influence of subsets of several variables on the target variable is not taken into account. Thus, several variables which individually have a small influence may together have a large influence on the target variable. To examine such subsets, multivariate feature filters (e.g. MANOVA) or wrapper methods must be used. These methods also take into account the interrelationship between the features.

In contrast to filter methods, wrapper methods use a specific model algorithm (model-specific). The results are therefore biased by the

selected model algorithm. The evaluation of the individual features is repeated for each subset, the generation of the subsets depends on the selected search strategy. The computation time of the wrapper methods is much higher than for the filter methods. The permutation FI is a wrapper method. Embedded methods create an evaluation of the feature selection while the model algorithm is executed. The methods are thus embedded in the learning algorithm. The computation time of embedded methods is lower than the computation time of wrapper methods [22]. An example of an embedded method is the random forest.

### 3.4. Model interpretability and explainability

The definition of *interpretability* in the context of ML systems according to Doshi-Velez and Kim [25] is *the ability to explain or to present in understandable terms to a human*. According to Brian and Cotton [26], interpretability is defined as follows, *systems are interpretable if their operations can be understood by a human, either through introspection or through a produced explanation*. Two types of interpretability were defined. The prediction interpretation, in which a justification for the prediction for a given prediction and usually non-interpretable models (e.g. neural networks) is made. The other variant is the interpretable models. In this variant, interpretable models are developed. Which are interpretable by themselves (introspection) and can be explained by reasoning.

In model interpretability, there is a trade-off between interpretability and performance. As model accuracy increases, model interpretability decreases [5]. For example, rule-based learning models or linear regressions have a high model interpretability but a low model accuracy. Whereas deep learning models have a low interpretability but a high model accuracy. For this reason, attempts to increase the interpretability of ML models should be made. The advantage of increased interpretability is, among other things, better implementability of ML models. When developing ML models, the implementability can be improved by considering interpretability as an additional design criterion for three reasons [5].

- Interpretability helps to provide impartial decision making, i.e., to detect and consequently correct biases in the training data.
- Interpretability makes it easier to provide robustness by highlighting potential adverse perturbations that could change the prediction.
- Interpretability can serve as an insurance that only meaningful variables infer the output, i.e., it guarantees that an underlying truthful causality exists in the modelling.

The *explainability*, as opposed to interpretability, is linked to the internal logic and mechanics within the machine learning model. The more explainable the model is, the better the human can understand why the internal processes happens while the model is training or making decisions [27]. Explainable models are interpretable by default but not always vice versa [28]. Explainability refers to reasons and details a model provides to make its operation clear or easy to understand. One of the main barriers to the use of ML in practice is the lack of explainability. For example, ML models often deliver good performance, but the path to the result is often not apparent. For this reason, the explainability of ML models must be considered. The goals of Explainable Artificial Intelligence (XAI) are trustworthiness, causality, transferability, informativeness, confidence, fairness, accessibility, interactivity and privacy awareness [5].

### 3.5. Permutation feature importance (PFI)

The Permutation Feature Importance (PFI) [29] describes the increase of the model error when a single feature value is randomly shuffled. Due the shuffling the correlation between the feature and the



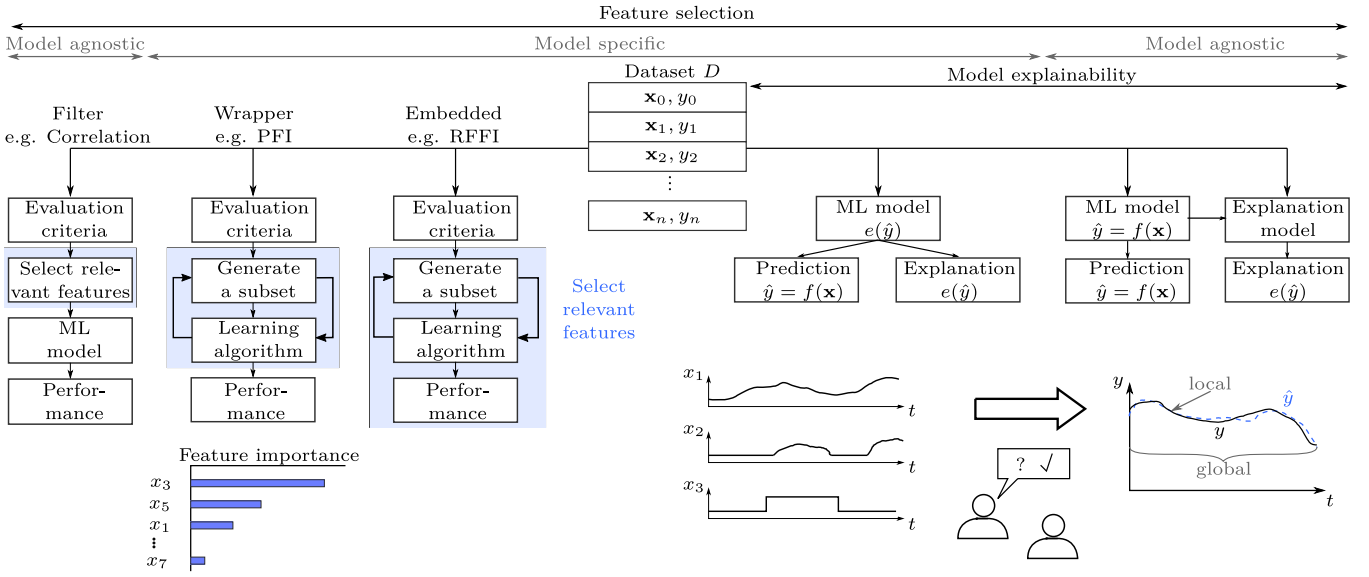


Fig. 1. Overview feature selection and model explanation (feature selection drawing inspired by: [23]).

label is removed. The larger the change in the label variable, the larger is the influence of the feature variable. Algorithm 1 shows the process of the PFI (based on [30]).

To calculate the PFI, the default model (without permutation)  $f$ , a data set  $D$  with features  $\mathbf{x}$  and label  $y$  is required. In the first step, the error for the standard model  $e_{\text{default}}$  is calculated. In this study, the negative mean square error (NMSE) is used as the error metric  $L$ . For each feature  $j$ , the values are shuffled and the whole is repeated for  $K$  repetitions. Shuffling the feature values resolves the relationship between the feature and the label. The model error  $e_{\text{perm}}$  is calculated for the permuted feature matrix  $\mathbf{X}_{\text{perm}}$ . The importance of a single feature  $\text{PFI}_j$  is calculated from the difference of the model error for the default model and the mean of the model error of the permuted data.

#### Algorithm 1 Permutation feature importance

**Require:** default model  $f$ , data set  $D$   
 Calculate the error of the original model  
 $e_{\text{default}} = L(y, \hat{f}(\mathbf{X}))$  (e.g. NMSE)  
**for** each feature  $j \in 1, \dots, p$ : **do**  
   **for** each repetition  $k \in 1, \dots, K$ : **do**  
     - Shuffle feature  $j$  which generates a permuted feature matrix  $\mathbf{X}_{\text{perm}}$ .  
     - Calculate the error  $e_{\text{perm}} = L(y, \hat{f}(\mathbf{X}_{\text{perm}}))$  with the permuted feature matrix  
   **end for**  
   Determine  $\text{PFI}_j = e_{\text{default}} - \frac{1}{K} \sum_{k=1}^K e_{\text{perm},j}$   
**end for**

### 3.6. Shapley additive explanations (SHAP)

Another method to determine the influence of a feature on a target value is to use SHAP values [31]. This game theoretic approach is used to justify the output of machine learning models and thus increase the transparency and interpretability of machine learning models. SHAP is a model-agnostic approach, so that the explanation can be considered separately from the model. Local and global explanations can be given for the model. The importance of the SHAP values is determined by the magnitude of the feature attributions, while the PFI is based on the reduction of the model accuracy. SHAP values are based on the Shapley values [32], which are a concept from cooperative game theory. The Shapley value  $\phi$  is a solution concept for the fair distribution of the payout in a cooperative game. This approach determines for coalitions

of players the importance of the players in each possible coalition. Some players have a higher importance on the total payout. Furthermore, the importance of the players is not only considered individually, but also in combination with other players. There is a set of players  $N = 1, \dots, n$ , where  $n$  is the number of players.  $S$  is a coalition of players. The characteristic function  $v$  assigns a real number to each coalition of players. This number is the value of a coalition  $v(S)$ , which represents the expected sum of the payout of the coalition, which is the total expected profit. The game is thus described by the tuple  $(v, N)$ .  $N \setminus \{i\}$  is all possible coalitions excluding the player  $i$ . The Shapley value  $\phi_i$  for player  $i \in C \subseteq N$  is given by Eq. (3).

$$\phi_i(N, v) = \sum_{S \subseteq N \setminus \{i\}} \frac{(n-1-|S|)! \cdot |S|!}{n!} \cdot \underbrace{(v(S \cup \{i\}) - v(S))}_{\text{marginal contribution}} \quad (3)$$

The Shapley value of the player  $i$  is the weighted marginal contribution of all possible coalitions. The player's marginal contribution is the difference between the value a coalition achieves with and without the player. Since the marginal contribution also depends on the order in which the players join, the calculation of  $\phi$  is done by considering all possible orders of the players. The weighting is done as follows. The number of possible combinations for  $n$  players is  $n!$ . For each coalition  $S \subseteq N \setminus \{i\}$  there are  $(n-1-|S|)! \cdot |S|!$  orders of precedence. There are  $|S|$  possibilities to arrange the players from  $S$  before the player  $i$  and  $(n-1-|S|)!$  possibilities to arrange the players from  $N \setminus \{S \cup \{i\}\}$  after the player  $i$ . Using Shapley values, the influence of different players on the payout and its fair distribution can be found. A fair payout can be considered to be the fulfilment of the properties efficiency, symmetry, dummy and additivity. The Shapley value is the only attribution method that fulfils all of these characteristics [30].

A variety of methods exist based on the Shapley value using different approaches [33]. The SHAP values [31] are based on the Shapley Values and have been extended thereby. Thus, SHAP combines the local interpretability of other model-agnostic approaches such as Local Interpretable Model-Agnostic Explanations (LIME) [34] and the game theory of Shapley values. This results in several properties in the form of local accuracy, missingness, consistency. Furthermore, SHAP included efficient methods for computing the Shapley values. The KernelSHAP is a kernel-based model, while the TreeSHAP is a tree-based model, both are used to estimate the Shapley values. In the scope of this work the TreeSHAP [35] is used to calculate the influence of each feature on the prediction. In the case of the SHAP values, the nomenclature is

**Table 1**  
Investigated room surfaces in the parameter study.

| Room type | $A_{\text{win}}$<br>m <sup>2</sup> | $A_{\text{wall}}$<br>m <sup>2</sup> | $A_{\text{win}}/A_{\text{wall}}$<br>1 | $A_{\text{wall}}/V_{\text{room}}$<br>m <sup>-1</sup> |
|-----------|------------------------------------|-------------------------------------|---------------------------------------|------------------------------------------------------|
| 1         | 7                                  | 3.5                                 | 0.67                                  | 0.2                                                  |
| 2         | 3.5                                | 7                                   | 0.33                                  | 0.2                                                  |
| 3         | 7                                  | 18.5                                | 0.27                                  | 0.49                                                 |

considered in the context of machine learning rather than game theory. So the players are single features or groups of features and the payout is the prediction of the model.

### 3.7. Justification for the choice of methods

This study analyses various methods for determining the FI and compares their characteristics and results. Different types of feature selection, the filter methods PCC, SRCC, MANOVA, the wrapper method PFI, the embedded method random forest feature importance (RFFI) and the model explainability method SHAP are investigated. These methods are used to find suitable features for ML prediction models. The filter methods are considered because there are fast and easy to use. The wrapper methods consider individual subsets of features, but are computationally intensive [3]. Through the elimination of iterative model training, XAI has computational advantages for feature selection. In addition, XAI methods provide insight into the model's decision-making process [36].

This model explainability offers great advantage over methods like filter, wrapper or embedded methods. For the explanation of black box models, LIME and SHAP are by far the most comprehensive and dominant methods in the literature for visualising feature importance and feature interactions [27]. It was shown that SHAP identify important features that have a major influence on the load forecast [37]. SHAP, which is the most commonly used model explainability method for power and energy applications [9], is considered in this work due to its frequent use and diverse possibilities.

### 3.8. Building description

The data set for the heating load prediction is generated by simulating a simplified room model. The room is represented by two RC elements. All exterior and interior components are combined into one equivalent element each. The floor area is 17.5 m<sup>2</sup> and the room has a south-facing exterior wall with a window. The modelling is based on the boundary conditions of test case 5 of VDI 6007-1 [38]. The consideration of the radiation through the transparent exterior components is carried out according to VDI 6007-2 [39]. The exact structure of the room model is described in Lauster et al. [40]. The modelling of the room is done with the modelling language *Modelica* in the modelling and simulation environment *Dymola*. The used room model is the thermal zone with two elements for exterior and interior walls model of the Buildings library version 8.0.0 [41].

For generating different room types, the exterior wall area  $A_{\text{wall}}$  and window area  $A_{\text{win}}$  were varied. Table 1 shows the surface properties of the room configurations. Room type 1 corresponds to the area values from VDI 6007. The window to wall ratio  $A_{\text{win}}/A_{\text{wall}}$  is 0.67 and the wall area to room volume ratio  $A_{\text{wall}}/V_{\text{room}}$  is 0.2. For room type 2, the areas for the exterior wall and window area were swapped. For the third variation, an interior wall was changed to an exterior wall.

These adjustments change the  $A_{\text{win}}/A_{\text{wall}}$  and the  $A_{\text{wall}}/V_{\text{room}}$  which have a large impact on the heat transmission and solar gains (Eq. (5) and (6)). Additionally to the wall areas, the specific heat capacity of the room was also varied. Light (50 Wh/m<sup>2</sup> K) and medium (90 Wh/m<sup>2</sup> K) thermal masses are studied. Besides the area properties of the room, the building physical properties  $U_{\text{wall}}$ ,  $U_{\text{win}}$ ,  $g$ -value of the window, as well as the usage properties like air exchange rate (AER) and internal

**Table 2**  
Investigated building characteristics in the parameter study.

| Nr. | $U_{\text{wall}}$<br>W/(m <sup>2</sup> ·K) | $U_{\text{win}}$<br>W/(m <sup>2</sup> ·K) | $g$ -value<br>1 | AER<br>h <sup>-1</sup> | $\dot{q}_{\text{IG}}$<br>W/m <sup>2</sup> |
|-----|--------------------------------------------|-------------------------------------------|-----------------|------------------------|-------------------------------------------|
| 1   | 0.15                                       | 0.8                                       | 0.15            | 0.25                   | 5.1                                       |
| 2   | 0.25                                       | 1.1                                       | 0.3             | 0.5                    | 10.3                                      |
| 3   | 0.5                                        | 1.4                                       | 0.45            | 0.75                   | 15.4                                      |
| 4   | 0.75                                       | 1.7                                       | 0.6             | 1                      | 20.6                                      |
| 5   | 1                                          | 2.1                                       | 0.75            | 2                      | 25.7                                      |
| 6   | 1.5                                        | 2.7                                       | 0.9             | 4                      | 30.9                                      |

gains were varied. This variation covers a large part of the building age classes, insulation types and types of use of office buildings. The ventilation and internal loads are applied from Monday to Friday between 7 am and 5 pm. The indoor room temperature in the model is ideally prescribed to be 20 °C in the heating case and 24 °C in the cooling case throughout the year. Within the room temperature band of 20–24 °C, there is neither heating nor cooling. The consideration period for the heating season is from 01. October to 31. April. If thermal losses are higher than gains, then heating to the temperature of 20 °C will occur in this period. The data set for training and testing the heating load prediction model uses this time period. As weather data the International Weather for Energy Calculations (IWECC) for the German cities Berlin and Stuttgart are used, which allows to consider different climatic zones. The weather data have a period of one year and a temporal resolution of one hour. To generate the data set, all room types, the thermal mass, building physical properties and usage properties are varied with each other. The  $U$ -values of the wall and the window are always considered in combination and are therefore not varied among each other, because they only have an influence on the average  $U$ -value of the exterior building components. Due to the variation of the properties, 15 552 variants exist.

### 3.9. Definition heating load and demand

The heating load  $\dot{Q}_{\text{heat}}$  of the building results from the difference of the gains and losses of the building (Eq. (4)). Gains are solar irradiance  $\dot{Q}_{\text{sol}}$  through the opaque components and internal gains  $\dot{Q}_{\text{IG}}$  which result from e.g. people, lighting and equipment. The losses are divided into transmission heat losses  $\dot{Q}_{\text{trans}}$  and ventilation heat losses  $\dot{Q}_{\text{vent}}$ . In addition, the building mass acts as a thermal storage and can store and discharge energy  $\dot{Q}_{\text{st}}$ .

$$\dot{Q}_{\text{heat}} = \underbrace{\dot{Q}_{\text{sol}} + \dot{Q}_{\text{IG}}}_{\text{gains}} + \underbrace{\dot{Q}_{\text{trans}} + \dot{Q}_{\text{vent}}}_{\text{losses}} + \dot{Q}_{\text{st}} \quad (4)$$

The  $\dot{Q}_{\text{trans}}$  is directly proportional to the thermal transmittance of the exterior components  $U_{\text{ext}}$  and the temperature difference  $\Delta T$  between the room temperature  $T_{\text{room}}$  and the outside temperature  $T_{\text{out}}$  (Eq. (5)). Besides the  $U$ -values of the exterior components, the A/V ratio and the share of window area also have a major influence on the heating load.

$$\dot{Q}_{\text{trans}} = U_{\text{ext}} \cdot A \cdot \Delta T \quad (5)$$

$$\dot{Q}_{\text{sol}} = g \cdot A_{\text{win}} \cdot I_{\text{sol,win}} \quad (6)$$

$\dot{Q}_{\text{vent}}$  has a directly proportional relationship with the temperature difference between  $T_{\text{room}}$  and  $T_{\text{out}}$  and the air exchange rate. The internal gains  $\dot{Q}_{\text{gain}}$  directly affect the room as convective and radiation heat, in the study the amount of gains is simplified based on the specific internal gains  $\dot{q}_{\text{IG}}$ . The solar gains  $\dot{Q}_{\text{sol}}$  depend on the window area  $A_{\text{win}}$ , the solar irradiance on the window  $I_{\text{sol,win}}$  and the  $g$ -value  $g$  of the window (Eq. (6)). The  $g$ -value indicates what proportion of radiation hitting the window is emitted by transmission or by convection – between the window and the room – to the room. The AER and  $\dot{q}_{\text{IG}}$  are equal to the values in Table 2 during the hours of use Monday to Friday from 7 am - 5 pm. Outside the hours of use, the air change and internal gains are 0.

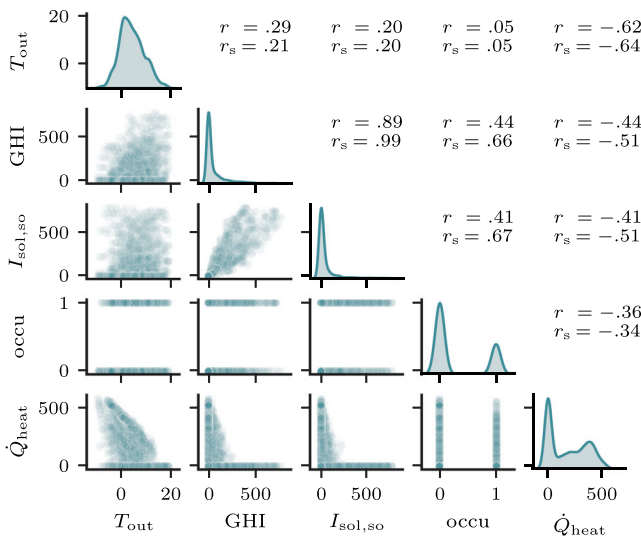


Fig. 2. Correlation between the features ( $T_{out}$ , GHI,  $I_{sol,so}$ , occu) and the label ( $\dot{Q}_{heat}$ ).

#### 4. Results

The results are divided into four parts. First, the results of the variation of the building properties and the resulting different data sets are presented in Section 4.1. For these data sets, the correlation between the features and the label in form of the heating load is presented (Section 4.2). The PFI and RFFI are shown in Section 4.3. The results from the FI for the SHAP values is described in Section 4.4.

##### 4.1. Building parameters study

For the example room, the location, the room types (Table 1), physical properties of building materials and the usage properties (Table 2) are varied among each other in a full factorial design. This results in a cartesian product with 15 552 room variants. The hourly feature values and heating and cooling loads for all room variants can be found in the GitHub repository [heating\\_cooling\\_load\\_VDI\\_6007](#).<sup>1</sup> Due to the large number of different variations, the maximum heating load  $\dot{Q}_{heat,max}$  varies widely. The maximum heating load is the highest heating load in the whole year. The range of  $\dot{Q}_{heat,max}$  for the different variations is between 0–4525 W (mean: 1340 W). For better comparability with other studies and for improved interpretation, the maximum heating load is also given in relation to the room area  $\dot{q}_{heat,max}$  is between 0–258.6 W m<sup>-2</sup> (mean: 76.6 W m<sup>-2</sup>). The relationship between the influencing variables and the heating load is examined in more detail using the default configuration defined here, room type 2, thermal mass medium,  $U$ -values 0.75 W/m<sup>2</sup> K and 1.7 W/m<sup>2</sup> K,  $g$ -value 0.45, AER 0.5 h<sup>-1</sup>,  $q_{IG}$  10.3 W/m<sup>2</sup> and the location is Berlin. In this configuration, the  $\dot{Q}_{heat,max}$  is 586 W. Fig. 2 shows the relationship between the features outdoor temperature  $T_{out}$ , global radiation GHI, specific solar irradiance on south-facing vertical surfaces  $I_{sol,so}$ , occupancy and the resulting heating power  $\dot{Q}_{heat}$ . Each point represents an hour. In the 5065 h of the period under consideration, a heating load was present in 3128 h. In many hours,  $\dot{Q}_{heat} = 0$  despite a  $T_{out}$  below 20 °C. This results from internal and solar gains. The values of  $T_{out}$  resemble a normal distribution, whereas the values of GHI and  $I_{sol,so}$  are an exponential distribution.

The PCC  $r$  and the SRCC  $r_s$  describe the correlation of the variables. The strongest correlation exists between  $\text{corr}(T_{out}, \dot{Q}_{heat})$  with  $-0.65$

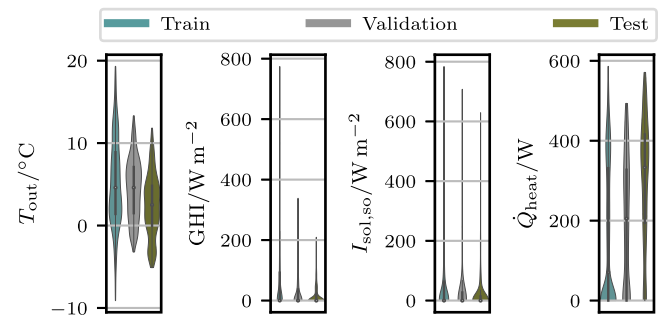


Fig. 3. Distribution of selected features and the label for the training, validation and test set.

and  $-0.67$ . A moderate correlation exists between  $\text{corr}(GHI, occu)$  and between  $\text{corr}(GHI, \dot{Q}_{heat})$ . The correlations between the features and the label are all negative. Therefore, an increasing value of the feature leads to a lower heating load. This is due to the fact that the features cause either heat gains (Eq. (4)) or a lower temperature gradient between indoor and outdoor and thus lower transmission and ventilation losses (Eq. (5)).

For training and testing the ML model, the data set is divided into training, validation and test sets. Only the heating period is considered. The time span of the training set is between 01.01–30.04 and 01.10–30.10 (dd.mm)(151 d, 71%). The validation set includes the period 01.11–30.11 (30 d, 14%) and the test set starts on 01.12 and ends on 31.12 (31 d, 15%). Fig. 3 shows the features  $T_{out}$ , GHI and the prediction variable  $\dot{Q}_{heat}$  for three different data sets. The binary value of occupancy was omitted from the presentation due to its lack of informational value. The values of the considered variables for the validation and test data are within the range of the training data. Thus, only predictions for individual values known from the training must be made. However, this does not mean that the combination of the individual features and the label was already used in the training. For comparability between the thermal balance of the building, the FI and the model explainability, only the data for the test set will be considered in the further course of the work. Otherwise, different time periods would be considered for the analyses, which could result in faulty comparability between the values.

To consider the thermal balance in the heating period, the energy quantities of heating, solar gains, internal gains, and losses in the form of transmission and ventilation are balanced over the entire heating period. In contrast to the maximum heating load, the heating energy demand and the thermal balance have a significantly larger influence for the annual consideration, since this represents all relevant scenarios and not only the maximum value. The absolute values of the heat demand  $Q_{heat}$  range between 0–1738 kWh (mean: 471 kWh) and the area-specific heat demand  $q_{heat}$  is between 0 – 99.3 kWh m<sup>-2</sup> (mean: 26.9 kWh m<sup>-2</sup>). The high solar and internal gains combined with low heat losses due to good insulation result in variants in which no heat demand is required in the month of December (test set). Furthermore, it should be noted that the rooms are cooled from a temperature of 24 °C and are heated up to 20 °C. In this way, the thermal mass can be loaded during the occupancy period and unloaded during the non-occupancy period. The heat demand for the different variants and thus also the parts of the thermal balance show a very high range. For better comparability, only the relative shares of the thermal balance are considered. For the default configuration, 56.9% are transmission losses  $Q_{trans}$  and 43.1% are ventilation heat losses  $Q_{ven}$ . The losses are covered by solar gains  $Q_{sol}$  27.2%, internal gains  $Q_{IG}$  16.4%, and heat from the heater  $Q_{heat}$  56.4%. Fig. 4 shows the distribution of thermal gains depending on the building characteristics.

The share of the heater  $Q_{heat}$  ranges from 0–98.0% (mean: 68.4%), the share of solar gains  $Q_{sol}$  ranges from 1.0–75.3% (mean: 14.7%)

<sup>1</sup> [GitHub repository](#)

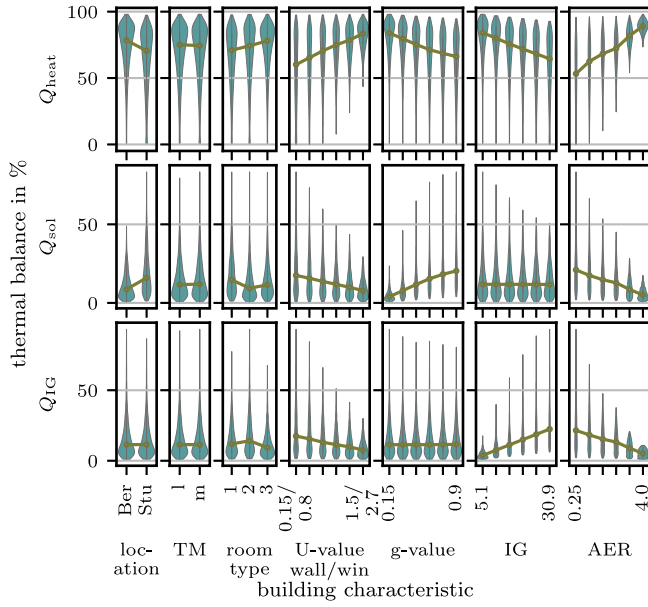


Fig. 4. Proportion of thermal balance depending on the building characteristics (olive line = course of the median values).

and internal gains  $Q_{IG}$  range from 1.0–86.4% (mean: 15.1%). The mean outside temperature  $T_{out}$  in Berlin with 2.42 °C is higher than in Stuttgart with 2.03 °C. The mean global radiation  $\overline{GHI}$ , on the other hand, is greater in Stuttgart with 29.1 W to 16.9 W than in Berlin. The mean heat demand  $\overline{Q}_{heat}$  for December in Stuttgart with 449.3 kWh is lower than that of Berlin with 492.9 kWh. The higher GHI thus have a greater influence on the  $Q_{heat}$  than the lower  $T_{out}$ . The share of  $Q_{IG}$  is higher in Stuttgart and in Berlin a higher share of the thermal balance has to be covered by  $Q_{heat}$  (Fig. 4). The thermal mass (TM) and the room type (RT) have a minor influence compared to the other building characteristics. A larger window area (RT 1 compared to RT 2) increases the mean proportion of solar gains. A larger proportion of external walls leads to a higher mean proportion of heating energy (RT 1/2 to RT 3). A lower  $U$ -value reduces the mean share of the heater and increases the share of solar and internal gains. A higher  $g$ -value or higher internal gains leads to a larger share of gains and thus to a reduction in the share of heating. An increasing AER, on the other hand, has a negative influence on the share of active heating, so the share  $Q_{heat}$  increases with an increasing AER. The results from the parameter study are plausible and correspond to the expected tendencies from the thermal balance (Eq. (4)).

To quantify the results of the thermal balance (Fig. 4), a MANOVA is performed between the building characteristics (independent variables) and the thermal balance (dependent variables). The MANOVA determines which building characteristics have a great influence on the thermal balance. Table 3 shows the results of the MANOVA. The python package statsmodels [42] was used to carry out the MANOVA. All building characteristics are statistically significant ( $p < 0.05$ ). A lower WL value means a greater discrimination ability of the building characteristics (independent variables) to the thermal balance (dependent variables). Thus, the thermal mass and the room type have a very low discriminant of  $WL = 0.9968$  and  $0.9313$ , respectively. The AER has the lowest WL with  $0.4263$  and the internal gain the second lowest value with  $0.4697$ . The two variables thus have a greater influence on the subdivision into individual groups of the thermal balance parts. The  $U$ -value and the  $g$ -value have a WL of  $0.5799$  and  $0.4913$ , respectively, indicating a medium discrimination of the dependent variable.

#### 4.2. Correlation between the features and the building characteristic

The relationship between the parts of the thermal balance and the correlations of the features and the heating load are examined. For better comparability with the FI and the model explainability, the correlation is determined exclusively for the test data set (December). According to Sun et al. [3], the features can be divided into meteorological information, indoor environmental information, occupancy related data, time index, building characteristic data, socio-economic information and historical data. The most frequently used features are meteorological, historical data and time index. Since the data sets are simulation data, only data that influence the results of the simulation are considered. This means, for example, that wind speed is excluded. Since only occupancies and internal loads of 0 or 100 % are present in the simulation and these are known, time index is not used. With the help of this time index, the occupancy could be learned from the training data. The correlation between GHI and  $I_{sol,so}$  is very strong ( $r = 0.88$ , see Fig. 2). This strong correlation can cause unrealistic data instances in the permutation feature importance, for example, if there is a very high GHI but no  $I_{sol,so}$ . Furthermore, adding a correlated feature can reduce the importance of the correlated feature, as the FI is split between both features [30,43]. For this reason, only  $I_{sol,so}$  is considered in the following considerations and not in combination with GHI.  $I_{sol,so}$  is chosen because it leads to a higher FI when considering GHI and  $I_{sol,so}$  separately. To reduce the number of evaluation parameters, only one correlation is evaluated in the following. Since the PCC is more for interval scale and the SCC more for ordinal scales, the PCC is chosen because more values have interval scale (Fig. 2).

For the thermal balance, the amount of energy is determined for the entire period of December and for the correlation, the hourly features and the label are used to determine the correlation. Since the energy shares cannot be used directly for the correlation, features that significantly influence the parts of the thermal balance are used for the correlation. The share of  $Q_{sol}$  is equated with  $I_{sol,so}$  and the share of  $Q_{IG}$  with occupancy. For the share of  $Q_{heat}$ ,  $T_{out}$  is used as the feature. Since this is the potential for the heat losses (Eq. (4)) in the form of transmission (Eq. (5)) and ventilation heat losses. Fig. 5 shows the PCC between the features and the heating load for all 15 552 variants depending on the building characteristics. Higher features – i.e. outdoor temperature, solar irradiance and occupancy – lead to a lower heating load. The individual correlations are thus negative. The correlation between the features and the heating load is strongly dependent on the building characteristics. The strongest correlation is on average between  $T_{out}$  and  $\dot{Q}_{heat}$  with values from  $-0.8990$  to  $0.1836$  (mean:  $-0.5495$ ). The sporadic positive correlation occurs with a low  $U$ -value, high internal loads and a low air exchange. In these variants, the heating load of the building is very small and the correlation between  $T_{out}$  and  $\dot{Q}_{heat}$  is not given due to the high internal loads and the low heat losses through transmission and ventilation.

With higher  $U$ -values and AER, the  $\text{corr}(T_{out}, \dot{Q}_{heat})$  increases. The proportion of internal gains in the thermal balance decreases and a larger proportion must be actively provided by a heater. The heating load is thus dependent on the outdoor temperature to a greater extent. The correlation trends of  $\text{corr}(T_{out}, \dot{Q}_{heat})$  for the  $U$ -values and the AER are contrary to those for the share of  $Q_{heat}$  in the thermal balance. Since a higher  $T_{out}$  leads to a lower share of  $\dot{Q}_{heat}$ . The  $\text{corr}(I_{sol,so}, \dot{Q}_{heat})$  has values ranging from  $-0.6522$  to  $-0.0134$  (mean:  $-0.4531$ ). The  $\text{corr}(T_{out}, \dot{Q}_{heat})$  and  $\text{corr}(I_{sol,so}, \dot{Q}_{heat})$  decreases with increasing  $g$ -values, because a higher  $g$ -value means that there are fewer hours of heating load. Higher  $U$ -values, on the other hand, lead to a stronger correlation, since heating load is also present at higher  $I_{sol,so}$  values. With a correlation of  $-0.7245$  to  $0.0676$  (mean:  $-0.4354$ ), occupancy has the lowest mean correlation of the shown features.

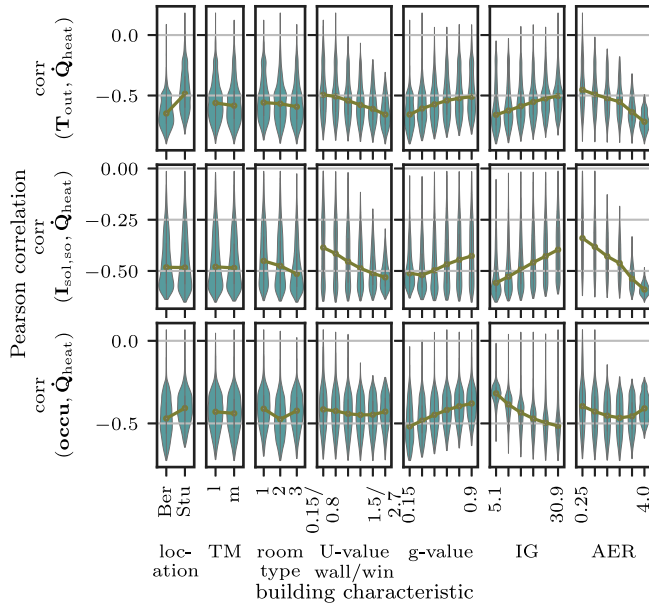
To quantify the influence of the building characteristics on the correlation, a MANOVA is carried out. All building characteristics show a statistical significance for the correlations. The thermal mass



**Table 3**

Results of the MANOVA to evaluate the influence of the building characteristics on thermal balance, PCC, PFI, RFFI, and SHAP values.

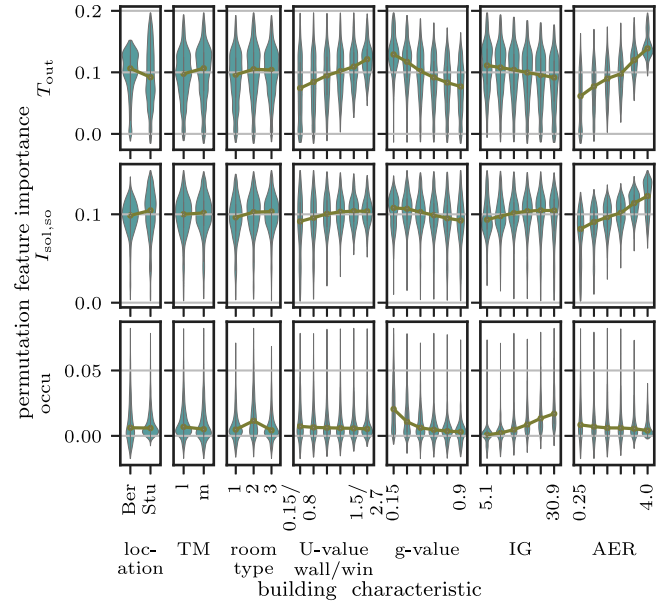
| Building characteristic | Thermal balance (Fig. 4) |      | PCC (Fig. 5) |      | PFI (Fig. 6) |      | RFFI   |      | relative SHAP value (Fig. 7) |      |
|-------------------------|--------------------------|------|--------------|------|--------------|------|--------|------|------------------------------|------|
|                         | WL                       | Rank | WL           | Rank | WL           | Rank | WL     | Rank | WL                           | Rank |
| location                | 0.6641                   | 3    | 0.2764       | 6    | 0.8901       | 3    | 0.9365 | 3    | 0.9408                       | 3    |
| thermal mass            | 0.9968                   | 1    | 0.9798       | 1    | 0.9755       | 1    | 0.9782 | 2    | 0.9789                       | 1    |
| room type               | 0.9313                   | 2    | 0.8644       | 2    | 0.9575       | 2    | 0.9928 | 1    | 0.9762                       | 2    |
| U-value                 | 0.5799                   | 4    | 0.5432       | 4    | 0.5272       | 5    | 0.8957 | 4    | 0.6264                       | 5    |
| g-value                 | 0.4913                   | 5    | 0.5954       | 3    | 0.3339       | 6    | 0.8957 | 4    | 0.5952                       | 6    |
| IG                      | 0.4697                   | 6    | 0.2952       | 5    | 0.5650       | 4    | 0.7076 | 7    | 0.7151                       | 4    |
| AER                     | 0.4263                   | 7    | 0.2673       | 7    | 0.3185       | 7    | 0.7536 | 6    | 0.5813                       | 7    |

**Fig. 5.** Pearson correlation coefficients between features and heating load depending on the building characteristics (olive line = course of the median values).

(WL = 0.9798) is the parameter with the lowest discrimination ability of the function and the location and the AER the parameters with the highest discrimination ability (WL = 0.2764 and 0.2673), followed by the internal gains with a WL of 0.2952. The ranking of the WL values for the various building characteristics is identical for the thermal balance and the PCC, for the thermal mass, room type, resistance and AER. Building characteristics that have a large or small influence on the thermal balance therefore tend to also have a large or small influence on the correlation between the features and the heating load.

#### 4.3. Importance of the features to predict the heating load

To determine the importance of each feature on the heating load, the two methods PFI (Section 3.5) and the RFFI are determined for each individual room variant. The models are trained with the training set and the PFI and RFFI are determined for the test set. The data sets were scaled to a value range of 0–1. Due to the dimensioned evaluation metric and the wide range of heating loads, all features and the label (heating load) of the 15552 variants were scaled for better comparability. The software library XGBoost [44] (eXtreme Gradient Boosting) in Python is used for the model fitting of the PFI. XGBoost is a distributed gradient boosting library that implements machine learning algorithms under the Gradient Boosting framework. XGBoost uses parallel tree boosting. The model has 100 gradient boosted trees. In PFI, the individual features are shuffled to see how this affects the model score. The NRMSE was used as the model score for the PFI. The PFI is executed with the Python library Scikit-learn [45]. The features were shuffled 50 times and the PFI for the features was determined for

**Fig. 6.** Permutation feature importance depending on the building characteristics (olive line = course of the median values).

each run. The average of these 50 runs was used for the evaluation. For the RFFI, the random forest regressor from Scikit-learn with 100 trees was used. For variants with a low number of heating hours, the model accuracy is in some cases poor, which means that the PFI and RFFI have little significance for these cases.

Fig. 6 shows the PFI for the features depending on the building characteristics. The lower the value, the lower the significance of the feature on the prediction. A negative value of the PFI means that the model with the shuffled feature has a better accuracy than without the shuffled feature. The highest mean PFI has  $I_{sol,so}$  with values between 0.002 – 0.149 (mean: 0.098). The  $T_{out}$  has an almost identical mean PFI with values of –0.015 – 0.196 (mean: 0.097). Occupancy has the lowest mean PFI with values ranging from –0.017 – 0.082 (mean: 0.010). For the RFFI, the mean decrease in impurity (MDI) is calculated. The sum of all FI is thus always 1. The highest RFFI has  $T_{out}$  with 0.44 – 0.93 (mean: 0.62), followed by the  $I_{sol,so}$  with 0.01 – 0.54 (mean: 0.35). The occupancy has very low values of 0.00 – 0.32 (mean: 0.04).  $T_{out}$  has a much higher mean value compared to the other features. The influence of  $T_{out}$  is thus higher for the RFFI than for the PCC and the PFI. For the RFFI and the PFI, occupancy is the feature with the lowest importance. The feature with the highest importance is  $I_{sol,so}$  for the PFI and  $T_{out}$  for the RFFI. In the work of Deng et al. [46] it was found that features with a high cardinality are preferred in RFFI. By permuting the features, this bias can be removed for features with high cardinality. This results in a low RFFI value for the occupancy, as the occupancy has only two values (low cardinality). Using PFI is therefore more appropriate.

The importance of the features for the model needs to be considered in the context of the building characteristics. The MANOVA shows

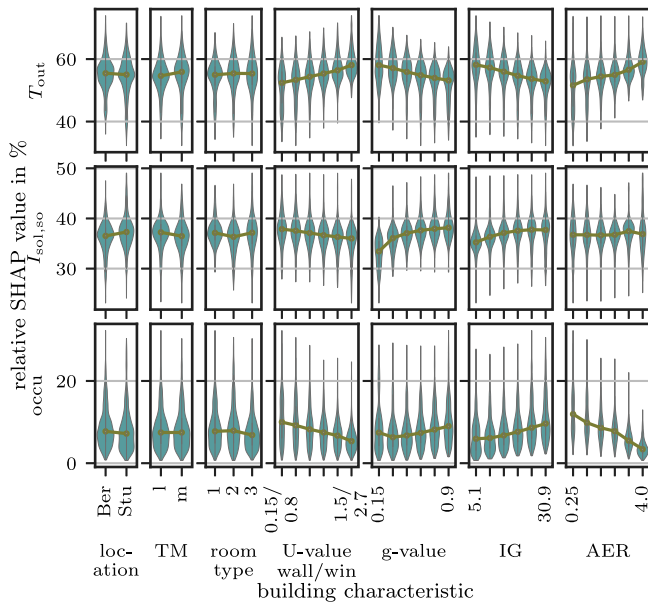


Fig. 7. Relative global SHAP value depending on the building characteristics (olive line = course of the median values).

the influence of the building characteristics for the PFI and the RFFI. The thermal mass and the room type have very small influence on the discreteness of the PFI and the RFFI. The AER has the greatest discrimination ability for the PFI and the internal load for the RFFI. The WL has lower values for the PFI than for the RFFI. The PFI thus shows a greater discrimination of the values depending on the building characteristics (see Table 3). The ranking of the WL values in the thermal balance and the PFI are similar. Based on the results, a higher proportion of the thermal balance generally leads to a greater FI. Furthermore, no general statement can be made about the importance of features in a heating load prediction, as the FI is highly dependent on the building characteristics.

#### 4.4. Determine model explainability with SHAP values

To determine the model explainability, the SHAP values of the features are determined. The influence of the features is not only considered separately, also the influence of the interrelationship between the features is considered. Fig. 7 shows the relative global SHAP values of the features depending on the building characteristics. Due to the different absolute SHAP values of the individual 15552 room variants and the resulting lack of comparability, the relative proportions of the SHAP values were determined for each variant. The relative SHAP value is the feature-specific proportion of the sum of all three SHAP values of  $T_{out}$ ,  $I_{sol,so}$  and occupancy.

The SHAP value is determined with the TreeSHAP method [35]. For this purpose, an XGBoost model with 100 gradient boosted trees is fitted and individual features are removed when determining the prediction value. The change in the predicted value is evaluated on the basis of the SHAP value. A negative SHAP value means that the value leads to a reduction of the prediction value, whereas a positive SHAP value leads to an increased prediction value. For the global SHAP value, the sum of all local absolute SHAP values is calculated. The feature  $T_{out}$  has the largest relative global SHAP values, which are between 32.3–73.9% (mean: 54.8%), followed by  $I_{sol,so}$  with 23.1–49.1% (mean: 36.8%). The occupancy 0.7–32.2% (mean: 8.5%) has lower proportions (see Fig. 7). The relative SHAP values for  $I_{sol,so}$  rise with increasing  $g$ -value. The same applies to the SHAP values of the occupancy with increasing internal loads. In general, SHAP values and PFI show similar trends for  $T_{out}$ , but diverge in certain cases for  $I_{sol,so}$  and occu (see

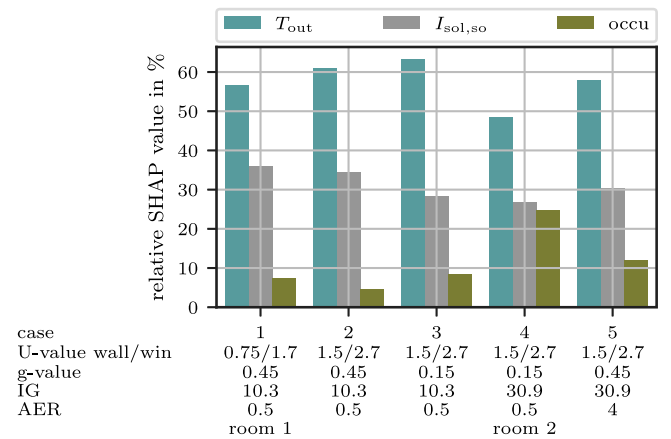


Fig. 8. Illustration of interaction effects of selected building characteristics based on case studies.

Figs. 7 and 6). The thermal mass and the room type lead to a low discrimination ability of the SHAP values, whereas the AER and the  $g$ -values lead to the highest discrimination (see Table 3).

Fig. 8 shows the relative SHAP values for five cases. This allows to show the interaction effects of building characteristics. Case 1 is the default room (room 1) with room type 2, thermal mass  $m$ ,  $U$ -values (wall:  $0.75 \text{ W/m}^2 \text{ K}$ , window:  $1.7 \text{ W/m}^2 \text{ K}$ ),  $g$ -value 0.45, AER  $0.5 \text{ h}^{-1}$  and  $q_{IG}$   $10.3 \text{ W/m}^2 \text{ K}$  at the location Berlin.

For each case, a building characteristic with high discriminatory ability of the SHAP values is varied. Poor insulation (case 2) results in a higher share of the SHAP value for  $T_{out}$ , while the share of the occupancy decreases. In case 3, the  $g$ -value of the windows is reduced, resulting in lower SHAP fraction of  $I_{sol,so}$ . Case 4 is also referred to as room 2. Which has the same characteristics as room 1 but poorer insulation, a lower  $g$ -value and a higher  $q_{IG}$ . The relative SHAP value of occu increases significantly and the SHAP value of  $T_{out}$  decreases due to the higher internal loads in case 4 compared to case 3. In case 5, the AER is changed from  $0.5 \text{ h}^{-1}$  to  $4 \text{ h}^{-1}$ . The SHAP value share of  $T_{out}$  is almost identical in case 5 (higher AER,  $U$ -value and IG) compared to case 1, despite the poorer insulation and the significantly higher AER. These two characteristics are counteracted by the higher internal gains. Therefore, all building characteristics must be considered holistically as they affect the SHAP values. For example, it can be seen that poorer insulation results in an on average higher relative SHAP value of  $T_{out}$  (see Fig. 8). However, this can be compensated by very high internal gains (see Fig. 8 case 1 and 5).

In addition to the global model explanation, SHAP values can also be used to carry out a local model explanation. The global evaluation is the sum of all absolute SHAP values, while the local evaluation shows the SHAP values for each time step. Fig. 9 shows the local SHAP values for room 1 and room 2. The upper subplot shows the scaled heating load.

The sum of the base value and the three SHAP values of  $T_{out}$ ,  $I_{sol,so}$  and occu at the respective time step is equal to the scaled heating load. Where the mean heating load from the training set is the base value. Subplot 2 and 3 show the local SHAP values for the default room (room 1) and the comparison room (room 2). The last subplot shows the scaled features ( $T_{out}$ ,  $I_{sol,so}$  and occupancy). The period of observation is from the 9th to the 11th of December and includes use and non-use days. The heating energy demand of room 1 over the period is  $13.8 \text{ kWh}$  and that of room 2 is  $23.1 \text{ kWh}$ . The heating load and thus SHAP values are scaled to a range of values from 0–1. This allows a better comparison of the variants with each other, as they have different maximum heating loads. The SHAP value indicates whether the feature leads to an increase or reduction compared to the mean heating load. The global SHAP values for the period from the

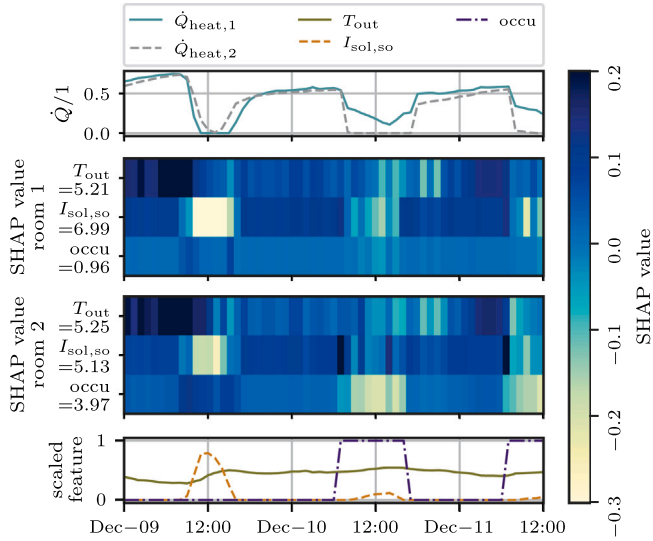


Fig. 9. Local SHAP values for two rooms depending on the feature values.

9th to the 11th of December are shown below the feature labels. Solar radiation occurs during the day on 09.12. This results in a negative SHAP value of  $I_{sol,so}$ , as solar radiation leads to a reduction in the heating load compared to the average heating load. In room 2, the global SHAP value from  $I_{sol,so}$  is lower because the  $g$ -value of the window is smaller. On the day of occupancy 10.12. The SHAP value of occupancy is reduced due to the presence of people and the associated internal load. In room 2, the internal load is three times higher than in room 1, which leads to a significantly lower SHAP value of occupancy. Lower  $T_{out}$ , as in the period from 5 am to 9 am on 9 December, result in high positive SHAP values of  $T_{out}$ , as these lead to an increase in the heating load. Overall, the global SHAP value for  $T_{out}$  is higher for room 1 than for room 2, despite the lower  $U$ -value. This is due to the interaction of all three SHAP values. Using the local SHAP values it is possible to observe the influence of the individual features on the SHAP values. The temporal course of the SHAP values can be used to identify different building characteristics such as insulation, occupancy and window characteristics.

For room 1 the calculation of the TreeSHAP was the fastest, with a computation time<sup>2</sup> for the standard room variant of 0.13 s, compared to the RFFI with 0.20 s and the PFI with 0.38 s. In contrast, the KernelSHAP computation time is 7.8 s.

For the final comparison of the influencing variables and features of the thermal balance, the correlation, the FI and the model explainability, all variables are normalised to a range of 0–1 (see Fig. 10). The normalisation of the values is carried out for each category (Share TB, Pearson, PFI, RFFI, relative global SHAP). The values  $Q_{heat}$  and the PCC  $T_{out}$  have similar normalised values. Whereas the correlations of the PCC  $I_{sol,so}$  have much higher values than  $Q_{sol}$ . The same counts for the occupancy and  $Q_{IG}$ . For PFI, the  $I_{sol,so}$  and the  $T_{out}$  have the highest mean FI. The RFFI and SHAP values for the individual features are in a similar range. Therefore, feature selection methods can sometimes produce varying results. It is recommended to use SHAP values because they provide local and global FI and other advantages.

It can be concluded that the  $T_{out}$  has the highest importance on the heating load prediction in most variants. Compared to  $I_{sol,so}$  and occupancy, the  $T_{out}$  probably has the highest prediction accuracy. If future features with a prediction error are used to predict the heating load, the heating load will also be incorrectly predicted. Buildings with good insulation and high occupancy or large windows therefore presumably

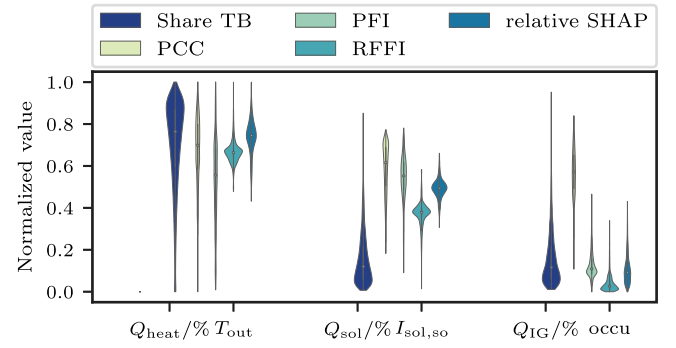


Fig. 10. Normalised values from thermal balance, PCC, PFI, RF, relative global SHAP value.

also have a higher relative prediction error when using future features than buildings with poor insulation and lower occupancy or smaller windows. In future work, it can be investigated how high the influence of the prediction error of the features is on the prediction accuracy of the heating load.

#### 4.5. Utilisation of the results for practical applications

Based on the SHAP values, buildings can be classified according to their building characteristics solely based on the input values of the features and the heating load. Additionally, it is possible to analyse for each time step which feature accounts for what proportion of the heating load. This provides precise information on the interaction of individual factors. Only white or grey box models of the building can provide this insight. Which requires a lot of information about the building. Local SHAP values can therefore be used to provide purely data-driven insights into the building, identifying ways to optimise it. By determining the influence of solar radiation and occupancy on the heating load, it is also possible to determine a normalised heating energy demand for different observation periods, independent of weather and occupancy. Furthermore, a hypothetical heating energy demand can be calculated by adjusting the feature values such as occupancy time or solar radiation.

## 5. Conclusion

This work investigated whether there is a systematic relationship between the importance of features in a heating load prediction and the building characteristics, and if so, how strong this relationship is. A simulation model was used to determine the hourly heating load of 15 552 different room variants with, e.g. different  $U$ -values and window areas. The thermal balance, PCC between features and heating load, RFFI, PFI and SHAP values were calculated and compared for these rooms. It was found that outside temperature is the most important feature for most variants. The FI of  $T_{out}$  increases with poorer insulation of the room and decreases with higher internal gains. Based on the simulations conducted, it was shown that the building characteristics of the room are implicitly present in the FI and that there is a relationship between the two values. These results can be used to estimate the FI values for the given room characteristics and serve as reference values. The AER and the  $g$ -value have the highest discrimination ability on the FI, while the thermal mass and the room type have a low value. The local SHAP values were used to determine the effect of the feature values on the heating load at each time step. This level of detail would otherwise only be possible using a white or grey box model of the building.

The accuracy of the model was low for room types with only a few hours of heating load due to the small amount of training data. Therefore, the FI is not necessarily representative for these variants.

<sup>2</sup> Performed with an AMD Ryzen 9 7950X 16-core processor.

The simulation models assumed some simplifications, in particular constant and regular occupancy. An important finding is that the FI must always be considered in the context of the given feature values. For example, the FI in April is different from the FI in December because the feature values and their occurrence are different in these months. By using the local SHAP values, these different feature values can be taken into account. In further investigations, the FI can be determined for real buildings and compared with the results of the simulation study. Moreover, it can be investigated if there is a relationship between the building characteristics and the FI of the cooling load.

### CRedit authorship contribution statement

**Alexander Neubauer:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Conceptualization. **Stefan Brandt:** Writing – review & editing. **Martin Kriegel:** Supervision, Resources.

### Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Data availability

The hourly heating load data sets for the 15552 room variants are available on [https://github.com/AlexanderNeubauer/heating\\_cooling\\_load\\_VDI\\_6007](https://github.com/AlexanderNeubauer/heating_cooling_load_VDI_6007).

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